Appendix B
Block Diagram Transformation Theorems

Complicated block diagrams can be broken into several easily recognized blocks. The summary of transformation theorems is a useful tool for this. W, X, Y, Z represent signals ***f(s)*** in the frequency domain. P, P1 and P2 represent transfer functions ***F(s)***.

Blocks in Cascade

$$Y = \left(P_1 P_2\right)\ \mathrm{X}$$

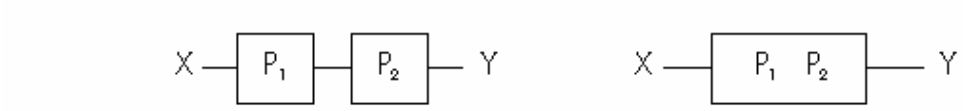

Figure B.1

Parallel Blocks

$$Y = P_1 X \pm P_2 X$$

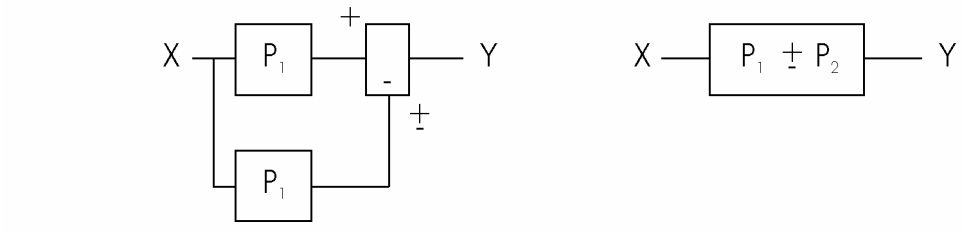

Figure B.2

Feedback Loop

$$Y = \frac{P_1}{P_1 X \pm P_2} X$$

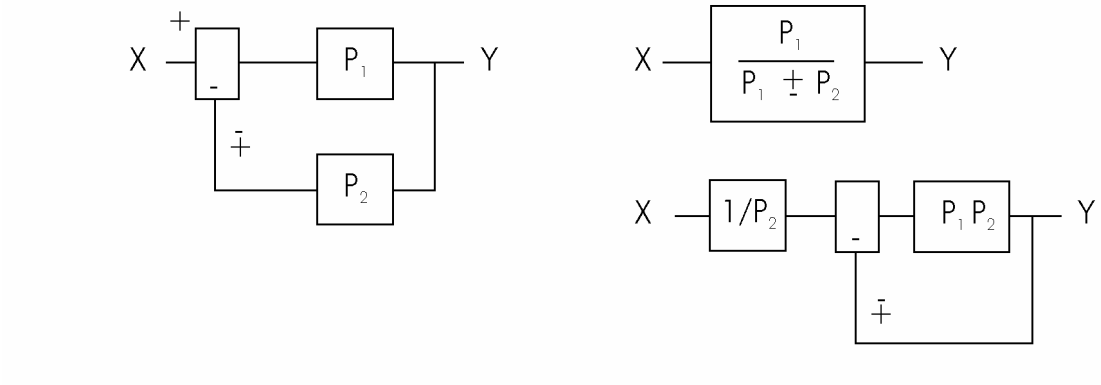

Figure B.3

Moving Summing Point

$$Z = P(X \pm Y)$$

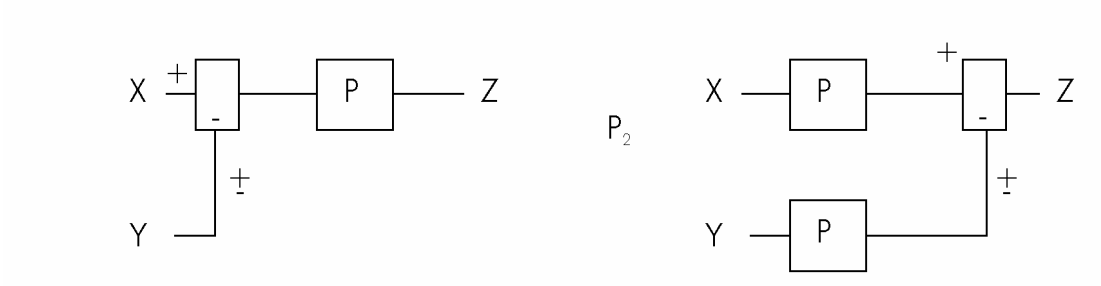

Figure B.4

Moving Takeoff Point

$$Y = PX$$

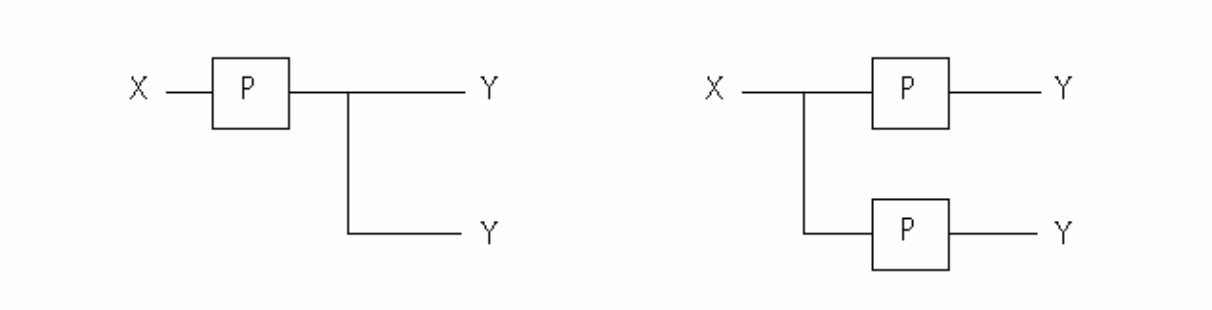

Figure B.5